

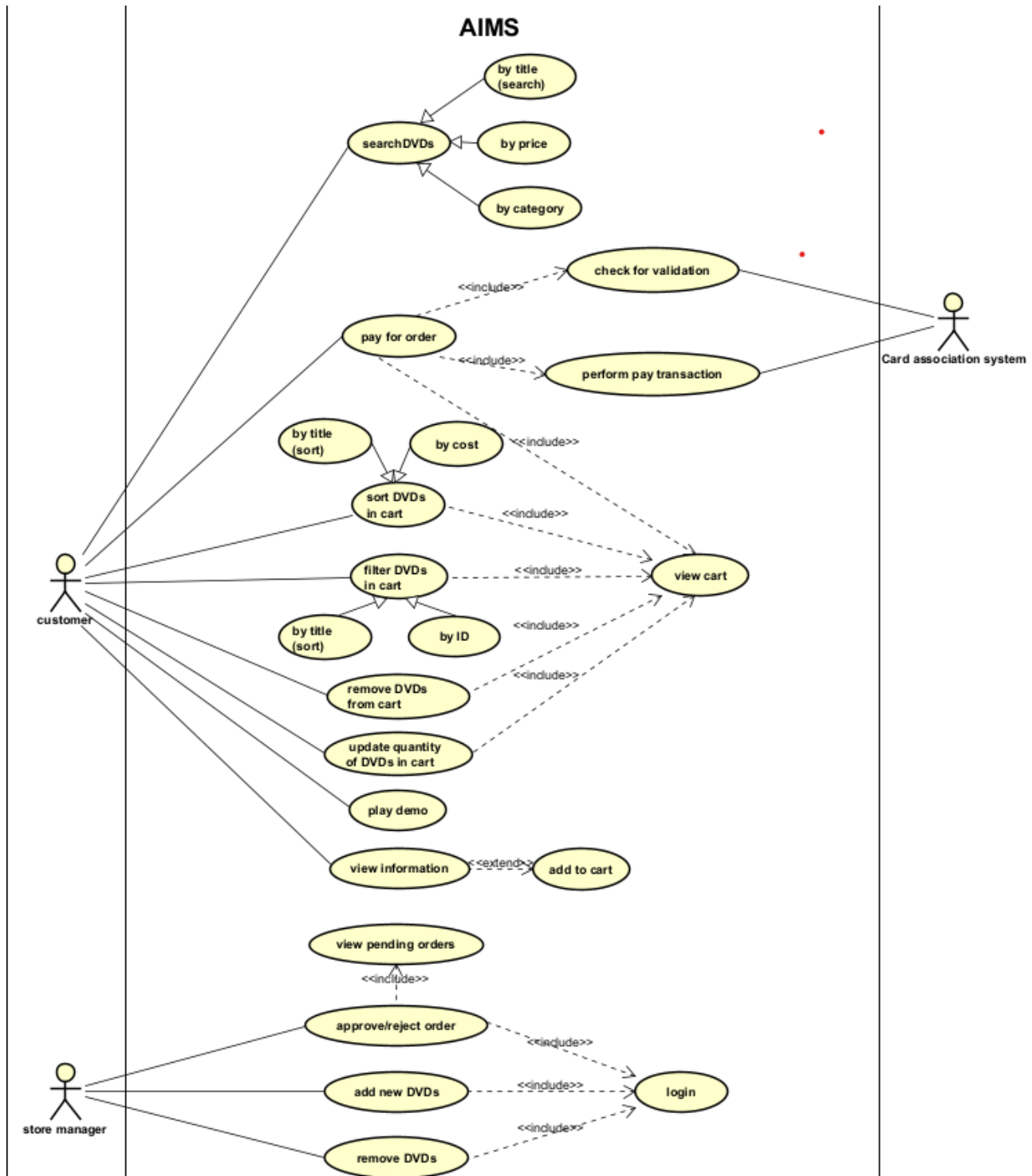
LAB 2. PROBLEM MODELLING AND ENCAPSULATION

IT3103E – 162065

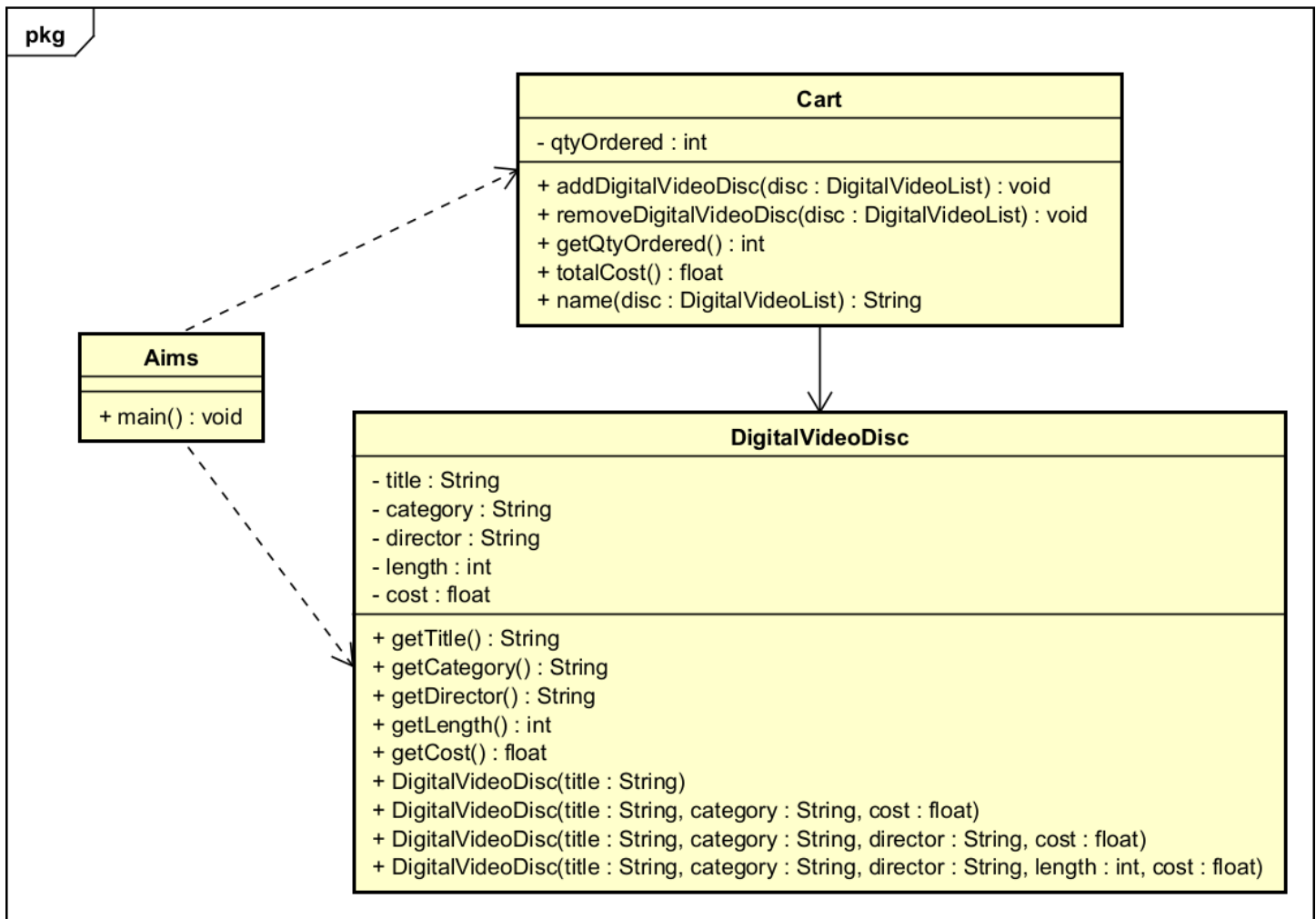
Họ và tên: Nguyễn Minh Tuấn

MSSV: 202416759

3. UseCase Diagram:



4. Class Diagram:



5. Class DigitalVideoDisc:

```
1 package Aims;
2
3 public class DigitalVideoDisc {
4     private String title;
5     private String category;
6     private String director;
7     private int length;
8     private float cost;
9     public String getTitle() {
10         return title;
11     }
12     public String getCategory() {
13         return category;
14     }
15     public String getDirector() {
16         return director;
17     }
18     public int getLength() {
19         return length;
20     }
21     public float getCost() {
22         return cost;
23     }
24     public DigitalVideoDisc(String title) {
25         super();
26         this.title = title;
27     }
28     public DigitalVideoDisc(String title, String category, float cost) {
29         super();
30         this.title = title;
31         this.category = category;
32         this.cost = cost;
33     }
34     public DigitalVideoDisc(String title, String category, String director, float cost) {
35         super();
36         this.title = title;
37         this.category = category;
38         this.director = director;
39         this.cost = cost;
40     }
41     public DigitalVideoDisc(String title, String category, String director, int length, float cost) {
42         super();
43         this.title = title;
44         this.category = category;
45         this.director = director;
46         this.length = length;
47         this.cost = cost;
48     }
49 }
```

6. Class Cart:

```
1 package Aims;
2
3 public class Cart {
4     public static final int MAX_NUMBERS_ORDERED = 20;
5     private DigitalVideoDisc itemsOrdered[] =
6         new DigitalVideoDisc[MAX_NUMBERS_ORDERED];
7     private int qtyOrdered = 0;
8     public void addDigitalVideoDisc(DigitalVideoDisc disc) {
9         if (qtyOrdered < MAX_NUMBERS_ORDERED) {
10             itemsOrdered[qtyOrdered] = disc;
11             qtyOrdered++;
12             System.out.println("The disc " + name(disc) + "has been added.");
13         } else {
14             System.out.println("The cart is almost full!");
15         }
16     }
17     public int getQtyOrdered() {
18         return qtyOrdered;
19     }
20     public String name(DigitalVideoDisc disc) {
21         return disc.getTitle();
22     }
23     public void removeDigitalVideoDisc(DigitalVideoDisc disc) {
24         for (int i = 0; i < qtyOrdered; i++) {
25             if (itemsOrdered[i] == disc) {
26                 // shift remaining items left
27                 for (int j = i; j < qtyOrdered - 1; j++) {
28                     itemsOrdered[j] = itemsOrdered[j + 1];
29                 }
30                 itemsOrdered[qtyOrdered - 1] = null; // clear last slot
31                 qtyOrdered--; // decrease count
32                 System.out.println("The disc " + name(disc) + "has been removed.");
33                 return;
34             }
35         }
36         System.out.println("Disc not found in the cart.");
37     }
38     public float totalCost() {
39         float total = 0;
40         for (int i = 0; i < qtyOrdered; i++) {
41             float price = itemsOrdered[i].getCost();
42             total += price;
43         }
44         return total;
45     }
46 }
47 }
```

7. Create Cart of DigitalVideoDiscs:

- Code:

```
1 package Aims;
2
3 public class Aims {
4
5     public static void main(String[] args) {
6
7         // Create a new cart
8         Cart anOrder = new Cart();
9
10        // Create new dvd objects and add them to the cart
11        DigitalVideoDisc dvd1 = new DigitalVideoDisc("The Lion King",
12            "Animation", "Roger Alfes", 87, 19.95f);
13        anOrder.addDigitalVideoDisc(dvd1);
14
15        DigitalVideoDisc dvd2 = new DigitalVideoDisc("Star Wars",
16            "Science Fiction", "George Lucas", 87, 24.95f);
17        anOrder.addDigitalVideoDisc(dvd2);
18
19        DigitalVideoDisc dvd3 = new DigitalVideoDisc("Aladin",
20            "Animation", 18.99f);
21        anOrder.addDigitalVideoDisc(dvd3);
22
23        // print total cost of the items in the cart
24        System.out.println("Total Cost is: ");
25        System.out.println(anOrder.totalCost());
26
27        // print total cost of the items in the cart after removing dvd2
28        anOrder.removeDigitalVideoDisc(dvd2);
29        System.out.println("Total Cost is: ");
30        System.out.println(anOrder.totalCost());
31    }
32 }
```

- Result:

```
The disc The Lion Kinghas been added.
The disc Star Warshas been added.
The disc Aladinhas been added.
Total Cost is:
63.89
The disc Star Warshas been removed.
Total Cost is:
38.940002
```

8. Reading Assignment:

